

Data cleaning and preparation

Week Two: Working with data

Outline for this week

- Week Two: data types, discovery, characterization, and assembly
- Discussion questions
- Homework assignment

Before class begins...

- Read Chapter 2 in your text.

Some data mining terminology

The datasets we will be working with look something like this:

This is a variable, attribute, or feature.

This is a value of the variable "MODEL."

ID	MPG	CYL	DISP	HP	WT	ACCEL	YR	ORIGIN	MODEL
1	18	8	307	130	3504	12	70	1	chevrolet chevelle malibu
2	15	8	350	165	3693	11.5	70	1	buick skylark 320
3	18	8	318	150	3436	11	70	1	plymouth satellite
4	16	8	304	150	3433	12	70	1	amc rebel sst
5	17	8	302	140	3449	10.5	70	1	ford torino
6	15	8	429	198	4341	10	70	1	ford galaxie 500
7	14	8	454	220	4354	9	70	1	chevrolet impala
8	14	8	440	215	4312	8.5	70	1	plymouth fury iii
9	14	8	455	225	4425	10	70	1	pontiac catalina
10	15	8	390	190	3850	8.5	70	1	amc ambassador dpl
11	NA	4	133	115	3090	17.5	70	2	citroen ds-21 pallas
12	NA	8	350	165	4142	11.5	70	1	chevrolet chevelle concours (sw)

A row of variable values is called an instance, record, or tuple.

Terms

Because data mining is a relatively new field, the terminology is still a little loose. “Variable,” “attribute,” and “feature” are often used interchangeably. Pyle defines a *feature* as relationships in the data—which implies that a feature is a combination of variables.

The *size* of a dataset is its width (number of variables) and depth (number of instances). The maximum number of unique values a variable can take is equal to the number of instances.

But what is a dataset, really?

A dataset is a representation of an object or system, characterized by the values, or measurements, of certain attributes. In this dataset snippet the object is a car. We study the relationships between the attributes so we can learn more about the car object. What are some of the relationships we can study in this dataset?

ID	MPG	CYL	DISP	HP	WT	ACCEL	YR	ORIGIN	MODEL
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What are some of the relationships we can study?

Suppose we wanted to determine the relationships (also called “learning the concept”) between the variables and miles per gallon. Then we would set mpg as our *target variable* or *class variable* and we would present the rest of the variables to the data mining algorithm (except for the ID field; we’ll get to that a bit later).

More on learning concepts

The “origin” field in this dataset is a coded variable where 1 = US, 2 = European, and 3 = Japanese.

Suppose we wanted to learn the concept “*Origin*” — we would set that variable as the target, and the algorithm would attempt to learn what distinguishes American cars from European and Japanese, etc.

Usually, a dataset has more than one possible target variable, which means we can look at the dataset from many different “views.”

Measurements

Attributes and their values are critically important for concept learning. As Pyle points out, there must be some validity to these measurements, or their usefulness in explaining relationships is limited or non-existent.

We use variability in the measurements to determine the validity of a particular attribute. What do we mean by variability? Where does it come from? Is it good or bad?

Variability

Variability can mean two things:

1)“Noise” in measurements: Noise is the inescapable “fuzz” around any measurement we make. We can minimize it, but we can never completely eliminate it. That’s “bad” variability.

2)A range of values in an attribute: Ranges of values are normally “good” variability. We want attributes to have different values, since a single-valued attribute tells us nothing. There’s such a thing as too many values, though.

Noise

We'll talk first about noise. It comes from several sources: the environment, the measuring device—even the person doing the measurement. It can be random or systematic. Let's do an experiment. Take your pulse for 10 seconds. Write down the count of heartbeats times six (the result is your pulse per minute). Wait five minutes (just sit there, don't run around the block), then take your pulse again. Repeat 5 times. Was the number always the same? Probably not. The difference in the numbers is random noise.

More on noise

So what is *systematic noise*? Suppose your calculator logic has a glitch, and instead of multiplying by the heartbeat count by 6 it multiplies by 5.5. The resulting numbers will differ from the “true” numbers by a fixed amount.

Noise in the physical world

Some of the noise in physical measures can come from the environment, from unit-to-unit differences, and from the resolution capacity of the measuring instrument. For instance, suppose you are collecting data on battery life; the temperature of the environment will have a large effect on the length a battery is usable (extreme hot or cold is detrimental to battery life).

Noise in the physical world

If I have 24 AA batteries, even if I can hold the environment constant, the unit-to-unit variation between individual batteries means the batteries will have different lifetimes. And, if I can only measure battery life in months, I won't have as much—or as accurate—information as I would if I could measure battery life down to the hour or minute. The resolution capacity is too low.

Another type of environmental noise

In order to mine a dataset, we usually have to “freeze” data collection at some point. Unless we are doing trend analysis, or time series mining, we want fairly static data. As we take snapshots of the “world” to capture the data we are interested in, the underlying environment changes somewhat. That change affects the relationships between variables slightly, and adds to the fuzziness of the measurement. It’s analogous to chaos theory—the end result is dependent on initial conditions.

Types of data

Not all data mining algorithms are capable of working with all data types. Forget for a bit your CS training—we aren't talking about double, long double, integer, or string here—we are talking about the basic characteristics of a measurement.

Scalar data types are point data—a single point or value defines the measurement. Within this type, we have several subtypes that supply different amounts and types of information. I'll just touch on some highlights, but be aware you are responsible for all the material in chapter 2.

Scalars

The first, least informational, scalar type is *nominal* data. We can see from the word *nominal* that this data is just *names*. Pyle distinguishes nominal data from categorical; in the literature you will see this type of data mostly called categorical. Be careful with categorical data represented as numbers. Don't make the mistake of treating them as numeric data points. For instance, the *Origin* attribute in the dataset snippet we saw earlier is a categorical variable. While we can take the average of the numbers 1 and 2, we can't take the average of "US" and "European."

A little more on categoricals

Categorical data such as SSN, zip code, and area code can be transformed to other features such as region or county. Zip codes have an associated latitude and longitude; area codes correspond to cities or counties; the three-digit prefix of social security numbers are mapped to the state where the person resided at the time the SSN was allocated.

A little more on categoricals

Another way to transform categoricals is to make them binary. Suppose we made the *Origin* attribute binary; we would then have three attributes (USA, Europe, Japan) instead of one.

Be aware that transforming these types of categorical values may, as in the case of the *Origin* example, increase the number of attributes.

Ordinals and intervals

Ordinal data is a very common type; we often see something like {small, medium, large}. However, we don't know the exact relationship between pairs—is a medium twice a small? Is a large three times a small? We don't have a reference scale to tell.

With interval data, we do know the differences between data points. We know 40 is twice 20. However, all temperature data does not have the same zero point, so we cannot say that the equivalent Celsius measures would have the same interval. We really cannot even say that 40 degrees F is twice as hot (what does that mean?) as 20 degrees F.

20 degrees F = -6 degrees C; 40 degrees F = 4 degrees C. It's clearly not the same size interval.

Special considerations

When we think about variability in the range of values, we must consider two extreme scenarios:

1. All instances of a variable have the **same** value. In this case, the variable cannot give any information about the dataset since its relationship with all other variables is constant. This variable (constant, really) should be removed.
2. Each instance of a variable has a **unique** value. In this case, the variable cannot give good information about the dataset because its relationship with all other variables is never the same. We should look carefully at the variable to see if we can group values together to reduce the number of unique measurements.

More special considerations

Monotonic variables are those whose values increase without bound, and that tend to show *interstitial linearity*, or a generally constant difference between two adjacent values. An example of a monotonic variable is a time or date stamp, or a record ID.

Sometimes we delete these variables from the dataset, but if we need them (in the case of times and dates, we may) we can transform them into interval data.

Sparse data occurs when most of the instances display no value for a particular variable. Depending on the importance of the variable, we may decide to keep it.

Outliers and anachronisms

An *outlier* is a data value far outside the range of most values of a particular attribute. Sometimes outliers are data entry errors, but sometimes they are valid values (as Pyle points out, insurance settlements are a good example). It's important to understand the domain so you know if the value is possible for that attribute.

Anachronisms are attributes that predict the class perfectly because they contain information that would not be present in new data that we would use for prediction. Can you think of an example?

Anachronisms should be removed.

Data integrity

Data that's been warehoused has gone through cleansing procedures that look for errors and inconsistencies. Many fielded data mining applications, however, don't use data warehouses. So, it's important to look for problems with data integrity.

What are data errors? Simply put, they are impossible or unacceptable values. For example, if you have an attribute named "color" then "sofa" is an unacceptable value. However, you may also encounter "vblue" –which is an easy enough error to fix; just replace with the most likely acceptable value, "blue". Fixing "sofa" may prove to be more difficult.

Data consistency

Refaat* defines data consistency as “consistent values in the same field over time” and a logical relationship between values in different fields.

What does that mean? I’ve been working with an injury dataset from the National Parks Department. We have one column that lists the type of injury, and another that captures the number of days of work an employee missed because of the injury. You would expect that the number of days has a logical relationship with the type of injury; a simple strain may result in 0-3 lost days, for example, while an employee who suffers a crush injury or leg fracture should be out of work for a substantial period of time.

Data consistency, part deux

However, this dataset is inconsistent—in more than half the instances, the number of days off is entered as zero regardless of the severity of the injury. In some instances, the number of days off is very large but the injury is a minor one.

Consistency within an attribute can be evaluated by looking at the range of values and their distribution. You would not expect air temperatures (on Earth, at least) to be greater than 200° F. You would also not expect negative numbers in a field tracking bank deposits.

What are some other consistency issues we should look for? The next slide contains some of Refaat's list.

Consistency issues

- Invalid characters
- Outliers
- Non-numeric data in a numeric field
- Missing or invalid row identifier
- Fixed values, such as 9999
- Unique records
- Date-triggered events
- Consistency of distributions between samples and full dataset, between scoring view and mining view.

Can you think of others?

Redundancies and constructed features

Mining algorithms are capable of learning quite complex concepts. However, we don't necessarily want them to “reinvent the wheel” and learn basic things like addition and subtraction. We can construct new attributes that are combinations of existing ones, and choose to allow redundancies that would assist the algorithm in learning more difficult relationships.

Recall the Time_In, Time_Out and Total_Time example in last week's lecture—Total_Time is a constructed feature. If we leave all three attributes in the dataset, the redundancy and new feature work to increase an algorithm's ability to learn more complex concepts by providing it with this basic relationship instead of requiring it to re-learn subtraction.

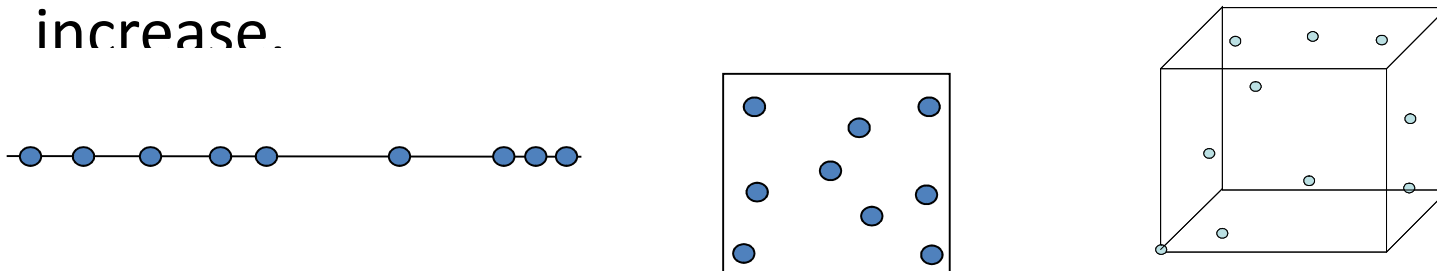
Adding to the width

Adding constructed features, such as Total_Time, or longitude and latitude instead of zip code, increases the width of the dataset and thus its size. Such changes have to be made judiciously, because increasing the size of a dataset—its *dimensionality*—has implications for the speed and accuracy of the mining task. The *curse of dimensionality* is a limiting factor on the feasibility of mining a large dataset.

The curse of dimensionality

Width, or the number of variables, is also called *dimension*. Typical data mining sets can have hundreds of variables and millions of instances. These multidimensional datasets have a volume of *width*depth*number of unique values*—which means the space that encloses the data is huge, and the data therefore becomes sparsely distributed throughout that space (mostly, along the edges).

We can see just moving from 1D to 3D how the data points move farther and farther from each other as the dimensions increase.



Alleviating the C of D

If you have 10 inputs that each take 10 possible values, you have 10^{10} possible different combinations of input values = 10,000,000,000 (10 billion). You would need 1 million instances to cover 0.01% of the possible patterns!

Judicious pruning of the dataset helps mitigate some, but not all, of this problem.

Summary

The data types we have in our datasets provide the information we need to learn concepts.

Obviously we would want data types that are rich in information, such as ratio data. We also want to minimize missing values and recognize variables that don't help the algorithm learn, such as constants and monotonic variables.

Assignments for this week

- Discussion questions:
 - I listed some aspects of data integrity and consistency that often arise in data mining projects. Can you think of other data integrity and consistency issues we should look for?
 - What do you suggest I do with the “number of days off” problem in the injury dataset?

Homework 1: due 2/10

- Download the HW1.csv file from MyCourses.
- Develop a data dictionary for this dataset. For each attribute:
 - Determine the data type
 - Determine the number of unique values
 - Determine the range of numeric values
 - Enumerate categorical or ordinal values
 - Note any inconsistencies you see in the dataset